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Lab 3: Multi-Source Retail Sales Data Integration and Analysis Objective

To import, clean, integrate, analyze, and store heterogeneous retail sales data from CSV, JSON, and Excel sources using R and SQLite.

Dataset

UCI Online Retail Dataset

The original dataset contains transaction-level retail sales information.

Data Sources Created transactions.csv — InvoiceNo, StockCode, CustomerID, Quantity, InvoiceDate products.json — StockCode, Description, UnitPrice customers.xlsx — CustomerID, Country

```
install.packages(c(
  "tidyverse",
  "jsonlite",
  "readxl",
  "writexl",
  "DBI",
  "RSQLite"
))
```

```
Installing packages into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)
```

```
library(tidyverse)
library(jsonlite)
library(readxl)
library(writexl)
library(DBI)
library(RSQLite)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
Attaching package: ‘jsonlite’


The following object is masked from ‘package:purrr’:

    flatten
```

Task 1: Data Import and Cleaning

The UCI Online Retail dataset was imported into R using read_excel().

The dataset was inspected for:

Missing values Duplicate records Invalid or zero quantities Invalid or zero unit prices Cleaning Decisions Records with missing CustomerID were removed because they could not be reliably associated with a customer. Records with missing Description were removed because product identification was incomplete. Duplicate records were removed using distinct(). Records with Quantity <= 0 were removed because they do not represent valid sales transactions. Records with UnitPrice <= 0 were removed because they cannot contribute to valid sales revenue. Revenue was calculated as:

Revenue = Quantity × UnitPrice

```
download.file(
  "https://archive.ics.uci.edu/static/public/352/online+retail.zip",
  "online_retail.zip"
)

unzip("online_retail.zip")

list.files()
```

```
[1] "Online Retail.xlsx" "online_retail.zip"  "sample_data"
```

```
retail_raw <- read_excel("Online Retail.xlsx")

head(retail_raw)
```

```
  InvoiceNo StockCode Description                         Quantity
1 536365    85123A    WHITE HANGING HEART T-LIGHT HOLDER  6
2 536365    71053     WHITE METAL LANTERN                 6
3 536365    84406B    CREAM CUPID HEARTS COAT HANGER      8
4 536365    84029G    KNITTED UNION FLAG HOT WATER BOTTLE 6
5 536365    84029E    RED WOOLLY HOTTIE WHITE HEART.      6
6 536365    22752     SET 7 BABUSHKA NESTING BOXES        2
  InvoiceDate         UnitPrice CustomerID Country
1 2010-12-01 08:26:00 2.55      17850      United Kingdom
2 2010-12-01 08:26:00 3.39      17850      United Kingdom
3 2010-12-01 08:26:00 2.75      17850      United Kingdom
```

```
4 2010-12-01 08:26:00 3.39      17850      United Kingdom
5 2010-12-01 08:26:00 3.39      17850      United Kingdom
6 2010-12-01 08:26:00 7.65      17850      United Kingdom

dim(retail_raw)

[1] 541909      8

str(retail_raw)

tibble [541,909 × 8] (S3: tbl_df/tbl/data.frame)
 $ InvoiceNo  : chr [1:541909] "536365" "536365" "536365" "536365" ...
 $ StockCode  : chr [1:541909] "85123A" "71053" "84406B" "84029G" ...
 $ Description: chr [1:541909] "WHITE HANGING HEART T-LIGHT HOLDER"
"WHITE METAL LANTERN" "CREAM CUPID HEARTS COAT HANGER" "KNITTED UNION
FLAG HOT WATER BOTTLE" ...
 $ Quantity   : num [1:541909] 6 6 8 6 6 2 6 6 6 32 ...
 $ InvoiceDate: POSIXct[1:541909], format: "2010-12-01 08:26:00"
"2010-12-01 08:26:00" ...
 $ UnitPrice  : num [1:541909] 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85
1.85 1.69 ...
 $ CustomerID : num [1:541909] 17850 17850 17850 17850 17850 ...
 $ Country    : chr [1:541909] "United Kingdom" "United Kingdom"
"United Kingdom" "United Kingdom" ...

summary(retail_raw)

     InvoiceNo          StockCode         Description         Quantity

 Length   :541909   Length   :541909   Length   :541909   Min.   :-
80995.000
 N.unique : 25900   N.unique :  4070   N.unique :  4211   1st Qu.:
1.000
 N.blank  :     0   N.blank  :     0   N.blank  :     0   Median :
3.000
 Min.nchar:     6   Min.nchar:     1   Min.nchar:     1   Mean   :
9.552
 Max.nchar:     7   Max.nchar:    12   Max.nchar:    35   3rd Qu.:
10.000
                                       NAs      :  1454   Max.   :
80995.000


  InvoiceDate                    UnitPrice            CustomerID
 Min.   :2010-12-01 08:26:00   Min.   :-11062.060   Min.   :12346
 1st Qu.:2011-03-28 11:34:00   1st Qu.:     1.250   1st Qu.:13953
 Median :2011-07-19 17:17:00   Median :     2.080   Median :15152
 Mean   :2011-07-04 13:34:57   Mean   :     4.611   Mean   :15288
 3rd Qu.:2011-10-19 11:27:00   3rd Qu.:     4.130   3rd Qu.:16791
 Max.   :2011-12-09 12:50:00   Max.   : 38970.000   Max.   :18287
                                                    NAs    :135080
```

```
      Country
 Length   :541909
 N.unique :    38
 N.blank  :     0
 Min.nchar:     3
 Max.nchar:    20
```

```
colSums(is.na(retail_raw))
```

```
  InvoiceNo   StockCode Description    Quantity InvoiceDate
UnitPrice
          0           0        1454           0           0
0
 CustomerID     Country
     135080           0
```

```
retail_clean <- retail_raw %>%
  filter(
    !is.na(Description),
    !is.na(CustomerID)
  )
```

```
dim(retail_clean)
```

```
[1] 406829      8
```

```
sum(duplicated(retail_clean))
```

```
[1] 5225
```

```
retail_clean <- retail_clean %>%
  distinct()
```

```
sum(duplicated(retail_clean))
```

```
[1] 0
```

```
dim(retail_clean)
```

```
[1] 401604      8
```

```
sum(retail_clean$Quantity <= 0)
```

```
[1] 8872
```

```
sum(retail_clean$Quantity == 0)
```

```
[1] 0
```

```r
sum(retail_clean$Quantity < 0)
```

```
[1] 8872
```

```r
retail_clean <- retail_clean %>%
  filter(Quantity > 0)

sum(retail_clean$Quantity <= 0)
```

```
[1] 0
```

```r
dim(retail_clean)
```

```
[1] 392732      8
```

```r
sum(retail_clean$UnitPrice <= 0)
```

```
[1] 40
```

```r
sum(retail_clean$UnitPrice == 0)
```

```
[1] 40
```

```r
retail_clean <- retail_clean %>%
  filter(UnitPrice > 0)

sum(retail_clean$UnitPrice <= 0)
```

```
[1] 0
```

```r
dim(retail_clean)
```

```
[1] 392692      8
```

```r
retail_clean <- retail_clean %>%
  filter(UnitPrice > 0)

sum(retail_clean$UnitPrice <= 0)
```

```
[1] 0
```

```r
dim(retail_clean)
```

```
[1] 392692      8
```

Cleaning Results Stage Records Original dataset 541,909

After missing-value removal 406,829

After duplicate removal 401,604

After invalid quantity removal 392,732

After invalid price removal 392,692

Final Cleaning Verification

Missing Description = 0

Missing CustomerID = 0

Duplicate records = 0

Invalid Quantity = 0

Invalid UnitPrice = 0

Final records = 392,692

```
list(
  Missing_Description = sum(is.na(retail_clean$Description)),
  Missing_CustomerID = sum(is.na(retail_clean$CustomerID)),
  Duplicate_Rows = sum(duplicated(retail_clean)),
  Invalid_Quantity = sum(retail_clean$Quantity <= 0),
  Invalid_UnitPrice = sum(retail_clean$UnitPrice <= 0),
  Final_Rows = nrow(retail_clean)
)
```

```
$Missing_Description
[1] 0

$Missing_CustomerID
[1] 0

$Duplicate_Rows
[1] 0

$Invalid_Quantity
[1] 0

$Invalid_UnitPrice
[1] 0

$Final_Rows
[1] 392692
```

```
transactions <- retail_clean %>%
  select(
    InvoiceNo,
    StockCode,
    CustomerID,
    Quantity,
    InvoiceDate
  )

head(transactions)
```

```
  InvoiceNo StockCode CustomerID Quantity InvoiceDate
1 536365    85123A    17850      6        2010-12-01 08:26:00
2 536365    71053     17850      6        2010-12-01 08:26:00
3 536365    84406B    17850      8        2010-12-01 08:26:00
4 536365    84029G    17850      6        2010-12-01 08:26:00
5 536365    84029E    17850      6        2010-12-01 08:26:00
6 536365    22752     17850      2        2010-12-01 08:26:00
```

```
dim(transactions)
```

```
[1] 392692      5
```

```
write_csv(transactions, "transactions.csv")
```

```
list.files()
```

```
[1] "Online Retail.xlsx" "online_retail.zip"  "sample_data"
[4] "transactions.csv"
```

```
products <- retail_clean %>%
  select(
    StockCode,
    Description,
    UnitPrice
  ) %>%
  distinct(StockCode, .keep_all = TRUE)
```

```
head(products)
```

```
  StockCode Description                          UnitPrice
1 85123A    WHITE HANGING HEART T-LIGHT HOLDER  2.55
2 71053     WHITE METAL LANTERN                 3.39
3 84406B    CREAM CUPID HEARTS COAT HANGER      2.75
4 84029G    KNITTED UNION FLAG HOT WATER BOTTLE 3.39
5 84029E    RED WOOLLY HOTTIE WHITE HEART.      3.39
6 22752     SET 7 BABUSHKA NESTING BOXES        7.65
```

```
dim(products)
```

```
[1] 3665    3
```

```
write_json(
  products,
  "products.json",
  pretty = TRUE,
  auto_unbox = TRUE
)
```

```
list.files()
```

```
[1] "Online Retail.xlsx" "online_retail.zip"  "products.json"
[4] "sample_data"        "transactions.csv"
```

Task 2: Data Integration

The three data sources were integrated using left_join().

Join Keys Transactions + Products → StockCode Transactions + Customers → CustomerID Join Justification

left_join() was selected because the transaction dataset is the primary dataset. It ensures that all valid transaction records are retained while matching product and customer information is added where available.

Integration Results

Transactions 392,692 5

Products 3,665 3

Customers 4,338 2

Final integrated dataset 392,692 10

Unmatched Records

Unmatched products = 0

Unmatched customers = 0

The final integrated dataset contains:

InvoiceNo, StockCode, CustomerID, Quantity, InvoiceDate, Description, UnitPrice, Revenue, Country, Customer_Category

```
customers <- retail_clean %>%
  select(
    CustomerID,
    Country
  ) %>%
  distinct(CustomerID, .keep_all = TRUE)

head(customers)
```

```
  CustomerID Country
1 17850      United Kingdom
2 13047      United Kingdom
3 12583      France
4 13748      United Kingdom
5 15100      United Kingdom
6 15291      United Kingdom
```

```
dim(customers)
```

```
[1] 4338    2
```

```r
write_xlsx(
  customers,
  "customers.xlsx"
)

list.files()
```

```
[1] "customers.xlsx"     "Online Retail.xlsx" "online_retail.zip"
[4] "products.json"      "sample_data"        "transactions.csv"
```

```r
transactions_imported <- read_csv("transactions.csv")
```

```
Rows: 392692 Columns: 5
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (1): StockCode
dbl  (3): InvoiceNo, CustomerID, Quantity
dttm (1): InvoiceDate

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet
this message.
```

```r
head(transactions_imported)
```

```
  InvoiceNo StockCode CustomerID Quantity InvoiceDate
1 536365    85123A    17850      6        2010-12-01 08:26:00
2 536365    71053     17850      6        2010-12-01 08:26:00
3 536365    84406B    17850      8        2010-12-01 08:26:00
4 536365    84029G    17850      6        2010-12-01 08:26:00
5 536365    84029E    17850      6        2010-12-01 08:26:00
6 536365    22752     17850      2        2010-12-01 08:26:00
```

```r
products_imported <- fromJSON("products.json")

head(products_imported)
```

```
  StockCode Description                          UnitPrice
1 85123A    WHITE HANGING HEART T-LIGHT HOLDER  2.55
2 71053     WHITE METAL LANTERN                 3.39
3 84406B    CREAM CUPID HEARTS COAT HANGER      2.75
4 84029G    KNITTED UNION FLAG HOT WATER BOTTLE 3.39
5 84029E    RED WOOLLY HOTTIE WHITE HEART.      3.39
6 22752     SET 7 BABUSHKA NESTING BOXES        7.65
```

```r
customers_imported <- read_excel("customers.xlsx")

head(customers_imported)
```

```
  CustomerID Country
1 17850      United Kingdom
```

```
2 13047      United Kingdom
3 12583      France
4 13748      United Kingdom
5 15100      United Kingdom
6 15291      United Kingdom
```

```
cat("Transactions:", dim(transactions_imported), "\n")
cat("Products:", dim(products_imported), "\n")
cat("Customers:", dim(customers_imported), "\n")
```

```
Transactions: 392692 5
Products: 3665 3
Customers: 4338 2
```

```
sales_data <- transactions_imported %>%
  left_join(
    products_imported,
    by = "StockCode"
  )
```

```
dim(sales_data)
```

```
[1] 392692      7
```

```
head(sales_data)
```

```
  InvoiceNo StockCode CustomerID Quantity InvoiceDate
1 536365    85123A    17850      6        2010-12-01 08:26:00
2 536365    71053     17850      6        2010-12-01 08:26:00
3 536365    84406B    17850      8        2010-12-01 08:26:00
4 536365    84029G    17850      6        2010-12-01 08:26:00
5 536365    84029E    17850      6        2010-12-01 08:26:00
6 536365    22752     17850      2        2010-12-01 08:26:00
  Description                         UnitPrice
1 WHITE HANGING HEART T-LIGHT HOLDER  2.55
2 WHITE METAL LANTERN                 3.39
3 CREAM CUPID HEARTS COAT HANGER      2.75
4 KNITTED UNION FLAG HOT WATER BOTTLE 3.39
5 RED WOOLLY HOTTIE WHITE HEART.      3.39
6 SET 7 BABUSHKA NESTING BOXES        7.65
```

```
sum(is.na(sales_data$Description))
```

```
[1] 0
```

```
sum(is.na(sales_data$UnitPrice))
```

```
[1] 0
```

```
sales_data <- sales_data %>%
  mutate(
```

```
    Revenue = Quantity * UnitPrice
  )

head(sales_data)
```

```
  InvoiceNo StockCode CustomerID Quantity InvoiceDate
1 536365    85123A    17850      6        2010-12-01 08:26:00
2 536365    71053     17850      6        2010-12-01 08:26:00
3 536365    84406B    17850      8        2010-12-01 08:26:00
4 536365    84029G    17850      6        2010-12-01 08:26:00
5 536365    84029E    17850      6        2010-12-01 08:26:00
6 536365    22752     17850      2        2010-12-01 08:26:00
  Description                         UnitPrice Revenue
1 WHITE HANGING HEART T-LIGHT HOLDER  2.55      15.30
2 WHITE METAL LANTERN                 3.39      20.34
3 CREAM CUPID HEARTS COAT HANGER      2.75      22.00
4 KNITTED UNION FLAG HOT WATER BOTTLE 3.39      20.34
5 RED WOOLLY HOTTIE WHITE HEART.      3.39      20.34
6 SET 7 BABUSHKA NESTING BOXES        7.65      15.30
```

```
sales_data %>%
  select(Quantity, UnitPrice, Revenue) %>%
  head()
```

```
  Quantity UnitPrice Revenue
1 6        2.55      15.30
2 6        3.39      20.34
3 8        2.75      22.00
4 6        3.39      20.34
5 6        3.39      20.34
6 2        7.65      15.30
```

```
final_data <- sales_data %>%
  left_join(
    customers_imported,
    by = "CustomerID"
  )

dim(final_data)
```

```
[1] 392692      9
```

```
head(final_data)
```

```
  InvoiceNo StockCode CustomerID Quantity InvoiceDate
1 536365    85123A    17850      6        2010-12-01 08:26:00
2 536365    71053     17850      6        2010-12-01 08:26:00
3 536365    84406B    17850      8        2010-12-01 08:26:00
4 536365    84029G    17850      6        2010-12-01 08:26:00
5 536365    84029E    17850      6        2010-12-01 08:26:00
6 536365    22752     17850      2        2010-12-01 08:26:00
```

```
  Description                          UnitPrice Revenue Country
1 WHITE HANGING HEART T-LIGHT HOLDER  2.55      15.30   United Kingdom
2 WHITE METAL LANTERN                 3.39      20.34   United Kingdom
3 CREAM CUPID HEARTS COAT HANGER      2.75      22.00   United Kingdom
4 KNITTED UNION FLAG HOT WATER BOTTLE 3.39      20.34   United Kingdom
5 RED WOOLLY HOTTIE WHITE HEART.      3.39      20.34   United Kingdom
6 SET 7 BABUSHKA NESTING BOXES        7.65      15.30   United Kingdom
```

```
sum(is.na(final_data$Country))
```

```
[1] 0
```

```
final_data %>%
  filter(is.na(Country)) %>%
  select(CustomerID) %>%
  distinct() %>%
  head()
```

```
     CustomerID
```

Task 3: Sales and Customer Analysis

3.1 Total Sales Revenue

The total revenue generated from the cleaned dataset is approximately:

9,546,219

3.2 Top 5 Products by Revenue Rank Product Revenue

1 PAPER CRAFT, LITTLE BIRDIE 168,469.60

2 REGENCY CAKESTAND 3 TIER 135,495.30

3 WHITE HANGING HEART T-LIGHT HOLDER 93,745.65

4 MEDIUM CERAMIC TOP STORAGE JAR 81,032.64

5 JUMBO BAG RED RETROSPOT 76,028.70

Observation: PAPER CRAFT, LITTLE BIRDIE generated the highest revenue among the products.

```
total_revenue <- final_data %>%
  summarise(
    Total_Revenue = sum(Revenue, na.rm = TRUE)
  )

total_revenue
```

```
  Total_Revenue
1 9546219
```

```
top_products <- final_data %>%
  group_by(StockCode, Description) %>%
  summarise(
    Total_Revenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(Total_Revenue)) %>%
  slice_head(n = 5)

top_products
```

```
  StockCode Description                          Total_Revenue
1 23843     PAPER CRAFT , LITTLE BIRDIE          168469.60
2 22423     REGENCY CAKESTAND 3 TIER             135495.30
3 85123A    WHITE HANGING HEART T-LIGHT HOLDER    93745.65
4 23166     MEDIUM CERAMIC TOP STORAGE JAR        81032.64
5 85099B    JUMBO BAG RED RETROSPOT               76028.70
```

```
top_countries <- final_data %>%
  group_by(Country) %>%
  summarise(
    Total_Revenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(Total_Revenue)) %>%
  slice_head(n = 5)
```

3.3 Top 5 Countries by Revenue Rank Country Revenue

1 United Kingdom 7,868,408.70

2 Netherlands 329,130.80

3 EIRE 287,260.90

4 Germany 233,804.10

5 France 203,628.30

Observation: The United Kingdom is the dominant market in terms of revenue.

```
top_countries
```

```
  Country        Total_Revenue
1 United Kingdom 7868408.7
2 Netherlands     329130.8
3 EIRE            287260.9
4 Germany         233804.1
5 France          203628.3
```

```
top_customers <- final_data %>%
  group_by(CustomerID) %>%
```

```
  summarise(
    Total_Purchase_Value = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(Total_Purchase_Value)) %>%
  slice_head(n = 5)
```

3.4 Top 5 Customers by Purchase Value Rank CustomerID Purchase Value

1 18102 383,153.00

2 14646 323,767.50

3 17450 171,051.90

4 16446 168,472.50

5 14911 155,092.00

Observation: Customer 18102 generated the highest total purchase value.

```
top_customers
```

```
  CustomerID Total_Purchase_Value
1 18102      383153.0
2 14646      323767.5
3 17450      171051.9
4 16446      168472.5
5 14911      155092.0
```

```
customer_value <- final_data %>%
  group_by(CustomerID) %>%
  summarise(
    Total_Purchase_Value = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  )

nrow(customer_value)
```

```
[1] 4338
```

```
quantile(
  customer_value$Total_Purchase_Value,
  probs = c(0.25, 0.50, 0.75),
  na.rm = TRUE
)
```

```
     25%      50%      75%
 317.835  703.570 1744.907
```

```
q1 <- 317.835
q2 <- 703.57
```

```
q3 <- 1744.9075

customer_value <- customer_value %>%
  mutate(
    Customer_Category = case_when(
      Total_Purchase_Value <= q1 ~ "Low Value",
      Total_Purchase_Value <= q2 ~ "Medium Value",
      Total_Purchase_Value <= q3 ~ "High Value",
      TRUE ~ "Premium"
    )
  )

customer_value %>%
  count(Customer_Category)
```

```
  Customer_Category n
1 High Value        1084
2 Low Value         1085
3 Medium Value      1084
4 Premium           1085
```

Customer Value Classification

Customers were classified using quartile-based thresholds calculated from total purchase values.

Category Threshold

Low Value ≤ 317.835

Medium Value 317.835 – 703.57

High Value 703.57 – 1,744.9075

Premium > 1,744.9075

The classification was implemented using the case_when() function.

Customer Distribution

Category Customers

Low Value 1,085

Medium Value 1,084

High Value 1,084

Premium 1,085

Total 4,338

```r
final_data <- final_data %>%
  left_join(
    customer_value %>%
      select(
        CustomerID,
        Customer_Category
      ),
    by = "CustomerID"
  )

head(final_data)
```

```
  InvoiceNo StockCode CustomerID Quantity InvoiceDate
1 536365    85123A    17850      6        2010-12-01 08:26:00
2 536365    71053     17850      6        2010-12-01 08:26:00
3 536365    84406B    17850      8        2010-12-01 08:26:00
4 536365    84029G    17850      6        2010-12-01 08:26:00
5 536365    84029E    17850      6        2010-12-01 08:26:00
6 536365    22752     17850      2        2010-12-01 08:26:00
  Description                         UnitPrice Revenue Country
1 WHITE HANGING HEART T-LIGHT HOLDER  2.55      15.30   United Kingdom
2 WHITE METAL LANTERN                 3.39      20.34   United Kingdom
3 CREAM CUPID HEARTS COAT HANGER      2.75      22.00   United Kingdom
4 KNITTED UNION FLAG HOT WATER BOTTLE 3.39      20.34   United Kingdom
5 RED WOOLLY HOTTIE WHITE HEART.      3.39      20.34   United Kingdom
6 SET 7 BABUSHKA NESTING BOXES        7.65      15.30   United Kingdom
  Customer_Category
1 Premium
2 Premium
3 Premium
4 Premium
5 Premium
6 Premium
```

```r
dim(final_data)
```

```
[1] 392692     10
```

```r
market_performance <- final_data %>%
  group_by(Country) %>%
  summarise(
    Total_Revenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(Total_Revenue))

head(market_performance, 5)
```

```
  Country        Total_Revenue
1 United Kingdom 7868408.7
2 Netherlands     329130.8
```

```
3 EIRE             287260.9
4 Germany          233804.1
5 France           203628.3
```

Market Performance High-Performing Market

United Kingdom

Revenue: 7,868,408.70

The United Kingdom is the strongest market and contributes substantially more revenue than the other countries.

Underperforming Market

Saudi Arabia

Revenue: 145.92

Saudi Arabia generated the lowest revenue among the countries in the dataset. The company could investigate demand, marketing reach, product availability, and distribution opportunities in this market.

```
tail(market_performance, 5)
```

```
  Country        Total_Revenue
1 Brazil         1184.43
2 RSA             971.82
3 Czech Republic  875.46
4 Bahrain         544.80
5 Saudi Arabia    145.92
```

Task 4: SQLite Database

The cleaned and integrated dataset was stored in a SQLite database:

Database: retail_sales.db

Table: retail_sales

The table contains:

392,692 records

10 columns

Database Columns

InvoiceNo

StockCode

CustomerID

Quantity

InvoiceDate

Description

UnitPrice

Revenue

Country

Customer_Category

SQLite Verification

The final dataset was successfully stored in the SQLite database retail_sales.db under the table retail_sales.

Database file exists: TRUE

Table name: retail_sales

Total records: 392,692

Total columns: 10

SQLite connection closed successfully after querying.

```
con <- dbConnect(
  SQLite(),
  "retail_sales.db"
)

dbIsValid(con)
```

```
[1] TRUE
```

```
dbWriteTable(
  con,
  "retail_sales",
  final_data,
  overwrite = TRUE
)

dbListTables(con)
```

```
[1] "retail_sales"
```

```
dbListFields(con, "retail_sales")
```

```
 [1] "InvoiceNo"         "StockCode"          "CustomerID"
 [4] "Quantity"          "InvoiceDate"        "Description"
 [7] "UnitPrice"         "Revenue"            "Country"
[10] "Customer_Category"
```

```
dbGetQuery(
  con,
  "SELECT COUNT(*) AS Total_Rows FROM retail_sales"
)
```

```
  Total_Rows
1 392692
```

```
query1 <- "
SELECT
    CustomerID,
    SUM(Revenue) AS Total_Revenue
FROM retail_sales
GROUP BY CustomerID
ORDER BY Total_Revenue DESC
LIMIT 5
"
```

SQL Query 1: Top 5 Customers by Revenue

The SQL query was executed from R using DBI and RSQLite.

```
top5_customers_sql <- dbGetQuery(con, query1)

top5_customers_sql
```

```
  CustomerID Total_Revenue
1 18102      383153.0
2 14646      323767.5
3 17450      171051.9
4 16446      168472.5
5 14911      155092.0
```

SQL Query 2: Revenue by Country

```
query2 <- "
SELECT
    Country,
    SUM(Revenue) AS Total_Revenue
FROM retail_sales
GROUP BY Country
ORDER BY Total_Revenue DESC
"

revenue_by_country_sql <- dbGetQuery(con, query2)

head(revenue_by_country_sql, 10)
```

```
   Country        Total_Revenue
1  United Kingdom 7868408.65
```

```
2  Netherlands     329130.77
3  EIRE            287260.87
4  Germany         233804.06
5  France          203628.28
6  Australia       157316.33
7  Spain            62080.85
8  Switzerland      57268.14
9  Belgium          43098.99
10 Japan            42082.88
```

```
dbDisconnect(con)
```

```
dbIsValid(con)
```

```
[1] FALSE
```

```
file.exists("retail_sales.db")
```

```
[1] TRUE
```

```
file.info("retail_sales.db")$size
```

```
[1] 37822464
```

Three Important Business Insights

1. United Kingdom is the dominant market

The United Kingdom generated approximately 7.87 million in revenue, making it the strongest market by a significant margin.

1. High-value customers and products contribute significantly to sales

Customer 18102 generated 383,153.00, while PAPER CRAFT, LITTLE BIRDIE generated 168,469.60. Identifying and retaining high-value customers and high-performing products can support revenue growth.

1. Market performance varies significantly

The United Kingdom generated approximately 7.87 million, whereas Saudi Arabia generated only 145.92. This large difference indicates substantial variation in market performance and highlights opportunities to investigate and improve weaker markets.

Conclusion

The heterogeneous retail datasets were successfully imported, cleaned, integrated, analyzed, and stored using R and SQLite. The final dataset contained 392,692 valid transaction records and 10 attributes. dplyr was used for data cleaning, joining, aggregation, and customer segmentation, while SQLite was used for persistent storage and SQL-based analysis. The analysis identified the company's top products, customers, and markets and provided business insights that can support retail decision-making.